

# Subnetting Made Simple

*A Beginner-Friendly Methodology with Practical Examples*

Subnetting can feel intimidating at first — but once you have a clear, repeatable process, it becomes second nature. This guide walks you through a straightforward 5-step methodology and two fully solved practice tasks. Whether you're preparing for a certification exam, a job interview, or simply want to sharpen your networking fundamentals, this method will help you work through any subnetting problem with confidence.

## The Methodology

When you need to further divide (subnet) an existing network, follow these five steps in order:

### Step 1 — Determine how many bits to add to the subnet mask

**Ask yourself:** what power of 2 equals the number of subnets I need?

Example: If you need **8 subnets**, find the exponent such that  $2^n = 8$ .

Since  $2^3 = 8$ , you need to add **3 bits** to the subnet mask.

### Step 2 — Update the prefix length

Add the number of bits found in Step 1 to your current prefix length.

Example: If your current prefix is `/25` and you're adding 3 bits:

$$25 + 3 = /28$$

### Step 3 — Calculate the number of IP addresses per subnet

Subtract the new prefix length from 32 to find the number of host bits remaining, then raise 2 to that power.

Example: With a new prefix of /28:

$$32 - 28 = 4 \rightarrow 2^4 = 16 \text{ addresses per subnet}$$

**⚠ Remember:** the first address is the **Network Address** and the last is the **Broadcast Address**. This leaves 14 usable host addresses.

#### Step 4 — Find the starting address (Network Address) of the nth subnet

Use the following formula:

$$\text{Start} = \text{Base Address} + \text{Subnet Size} \times (n - 1)$$

The **(n - 1)** is key: the first subnet already starts at offset 0, so you only need to jump forward **(n - 1)** blocks to reach the nth subnet.

**If the result exceeds 255 in the last octet, carry over into the next octet:**

$$\text{If result} > 255: \text{ next octet} += \text{floor}(\text{result} / 256) \quad \text{remainder} = \text{result} \bmod 256$$

#### Step 5 — Find the last usable host

Once you know the subnet's starting address:

$$\text{Last Usable Host} = \text{Network Address} + \text{Subnet Size} - 2$$

**We subtract 2 because:** 1 address is reserved for the Network Address (first), and 1 for the Broadcast Address (last). Everything in between is usable.

## Quick Reference

| What to Find | Formula                                                            |
|--------------|--------------------------------------------------------------------|
| Bits to add  | $2^n = \text{number of subnets} \rightarrow n \text{ bits to add}$ |

|                  |                                |
|------------------|--------------------------------|
| New prefix       | Old prefix + n                 |
| Subnet size      | $2^{(32 - \text{new prefix})}$ |
| Subnet start     | Base + Size $\times$ (n - 1)   |
| Last usable host | Subnet start + Size - 2        |

# Solved Practice Tasks

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## Task 1

|          |                                                 |
|----------|-------------------------------------------------|
| Split    | 192.168.0.0/23                                  |
| Into     | 8 subnets                                       |
| Question | What is the last usable host in the 6th subnet? |

## Solution

1. Find bits to add:  $8 = 2^3 \rightarrow$  add **3 bits**
2. Update prefix:  $23 + 3 = /26 \rightarrow$  **26 bits** used for the network part
3. Find host bits:  $32 - 26 = 6 \rightarrow$  **6 bits** for hosts  $\rightarrow 2^6 = 64$  addresses per subnet
4. Find start of 6th subnet:  $192.168.0.0 + 64 \times (6 - 1) = 64 \times 5 = 320$
5. Apply carry-over:  $320 > 255 \rightarrow 320 - 256 = 64$ , increment 3rd octet by 1  $\rightarrow$  **192.168.1.64**
6. Last usable host:  $192.168.1.64 + 64 - 2 = 192.168.1.126$

✓ The last usable host in the 6th subnet is **192.168.1.126**

## Task 2

|          |                                                 |
|----------|-------------------------------------------------|
| Split    | 10.0.0.0/23                                     |
| Into     | 4 subnets                                       |
| Question | What is the last usable host in the 3rd subnet? |

## Solution

7. Find bits to add:  $4 = 2^2 \rightarrow$  add **2 bits**
8. Update prefix:  $23 + 2 = /25 \rightarrow$  **25 bits** used for the network part
9. Find host bits:  $32 - 25 = 7 \rightarrow$  **7 bits** for hosts  $\rightarrow 2^7 = 128$  addresses per subnet
10. Find start of 3rd subnet:  $10.0.0.0 + 128 \times (3 - 1) = 128 \times 2 = 256$
11. Apply carry-over:  $256 = 256 \times 1 + 0 \rightarrow$  3rd octet +1, 4th octet = 0  $\rightarrow$  **10.0.1.0** (each octet holds values 0–255, i.e. 256 possible values, so anything  $\geq 256$  overflows into the next octet)

12. Last usable host:  $10.0.1.0 + 128 - 2 = 10.0.1.126$

✓ The last usable host in the 3rd subnet is 10.0.1.126

**Pro tip:** The  $(n - 1)$  in the formula is easy to forget. Remember: the 1st subnet starts at offset 0, the 2nd at offset 1, the 3rd at offset 2 — you always lag behind by one. Think of it like floors in a building: ground floor is Floor 1, but you climbed 0 staircases to get there.